

INTEGRALS OF THE FORM

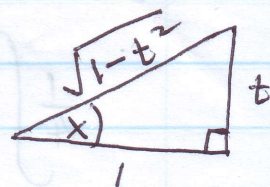
$$\int \frac{1}{a+b\sin^2 x + c\cos^2 x} dx$$

Lesson 9

Given $\int \frac{1}{a+b\sin^2 x + c\cos^2 x} dx$

let $t = \tan x$

$$dt = \sec^2 x dx$$



Notice the
square of Sine & Cosine

$$\frac{dx}{dt} = \frac{1}{\sec^2 x} = \frac{1}{1+\tan^2 x} \quad \therefore \sin x = \frac{t}{\sqrt{1+t^2}}$$

Thus you let

$$\therefore \frac{dx}{dt} = \frac{1}{1+t^2} \quad \therefore \cos x = \frac{1}{\sqrt{1+t^2}}$$

$$t = \tan x$$

Substitute: $dx = \frac{1}{1+t^2} dt$

$$\sin x = \frac{t}{\sqrt{1+t^2}}$$

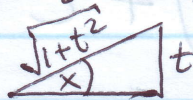
$$\cos x = \frac{1}{\sqrt{1+t^2}}$$

Example 1

Integrate w.r.t x , $\int \frac{1}{3+\cos^2 x} dx$

Solution

Let $t = \tan x$, $dx = \frac{1}{1+t^2} dt$, $\cos x = \frac{1}{\sqrt{1+t^2}}$



Substitute:

$$3 + \cos^2 x = 3 + \frac{1}{1+t^2} = \frac{3+3t^2+1}{1+t^2} = \frac{4+3t^2}{1+t^2}$$

So that:

$$\frac{1}{3+\cos^2 x} dx = \frac{1+t^2}{4+3t^2} \cdot \frac{1}{1+t^2} dt = \frac{1}{4+3t^2} dt$$

$$\int \frac{1}{3+\cos^2 x} dx = \int \frac{1}{4+3t^2} dt = \frac{1}{4} \int \frac{1}{1 + \left(\frac{\sqrt{3}}{2}t\right)^2} dt, \text{ let } u = \frac{\sqrt{3}}{2}t$$

$$du = \frac{\sqrt{3}}{2} dt$$

$$\therefore \frac{2}{\sqrt{3}} du = dt$$

Substitute:

$$\frac{1}{4} \int \frac{1}{1 + \left(\frac{\sqrt{3}}{2}t\right)^2} dt = \frac{1}{4} \int \frac{1}{1 + u^2} \cdot \frac{2}{\sqrt{3}} du$$

$$= \frac{1}{4} \cdot \frac{2}{\sqrt{3}} \int \frac{1}{1 + u^2} du$$

$$= \frac{1}{2\sqrt{3}} \tan^{-1} u + C$$

$$= \frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}}{2} t \right) + C$$

$$= \frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}}{2} \tan x \right) + C$$

but

$$t = \tan x$$

Exercise: (try the following)

Evaluate the integral $\int \frac{1}{2\cos^2 x + 1} dx$

Ans:

$$\int \frac{1}{2\cos^2 x + 1} dx = \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{1}{\sqrt{3}} \tan x \right) + C$$

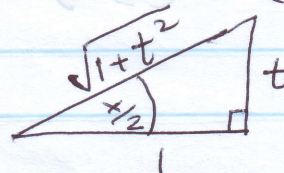
INTEGRALS OF THE FORM: $\int \frac{1}{a+b\sin x + c\cos x} dx$

Given $\int \frac{1}{a+b\sin x + c\cos x} dx$

Notice there are no squares here

let $t = \tan\left(\frac{x}{2}\right)$

Notice the half angle



You let $t = \tan\left(\frac{x}{2}\right)$

Substitute:

$$dx = \frac{2}{1+t^2} dt$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$dx = \frac{2}{1+t^2} dt$$

$$\cos\left(\frac{x}{2}\right) = \frac{1}{\sqrt{1+t^2}}$$

$$\sin\left(\frac{x}{2}\right) = \frac{t}{\sqrt{1+t^2}}$$

By double angle properties

$$\sin x = 2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)$$

$$\therefore \sin x = \frac{2t}{1+t^2}$$

$$\therefore \cos x = \frac{1-t^2}{1+t^2}$$

Example 2

Integrate w.r.t x , $\int \frac{1}{5+4\cos x} dx$

Solution

let $t = \tan\left(\frac{x}{2}\right)$, $\cos x = \frac{1-t^2}{1+t^2}$

$$dx = \frac{2}{1+t^2} dt$$

Substitute:

$$5+4\cos x = 5 + \frac{4(1-t^2)}{1+t^2} = \frac{5+5t^2+4-4t^2}{1+t^2} = \frac{9+5t^2}{1+t^2}$$

So that:

$$\frac{1}{5+4\cos x} dx = \frac{1+t^2}{9+5t^2} \cdot \frac{2}{1+t^2} dt = \frac{2}{9+5t^2} dt$$

$$\int \frac{1}{5+4\cos x} dx = \int \frac{2}{9+5t^2} dt = \frac{2}{9} \int \frac{1}{1+\left(\frac{\sqrt{5}}{3}t\right)^2} dt$$

let $u = \frac{\sqrt{5}}{3}t$
 $du = \frac{\sqrt{5}}{3} dt$
 $\therefore \frac{3}{\sqrt{5}} du = dt$

Substitute:

$$\frac{2}{9} \int \frac{1}{1 + \left(\frac{\sqrt{5}}{3}t\right)^2} dt = \frac{2}{9} \int \frac{1}{1 + u^2} \cdot \frac{3}{\sqrt{5}} du$$

$$= \frac{2}{9} \cdot \frac{3}{\sqrt{5}} \int \frac{1}{1 + u^2} du$$

$$= \frac{2}{3\sqrt{5}} \tan^{-1}(u) + C$$

$$= \frac{2}{3\sqrt{5}} \tan^{-1}\left(\frac{\sqrt{5}}{3}t\right) + C, \quad \text{but } t = \tan\left(\frac{x}{2}\right)$$

$$= \frac{2}{3\sqrt{5}} \tan^{-1}\left[\frac{\sqrt{5}}{3} \tan\left(\frac{x}{2}\right)\right] + C$$

Exercise (try:)

Evaluate the integral: $\int \frac{1}{3\sin x + 4\cos x} dx$

Answer:

$$\int \frac{1}{3\sin x + 4\cos x} dx = \frac{1}{5} \ln \left[\frac{1 + 2 \tan\left(\frac{x}{2}\right)}{4 - 2 \tan\left(\frac{x}{2}\right)} \right] + C$$
